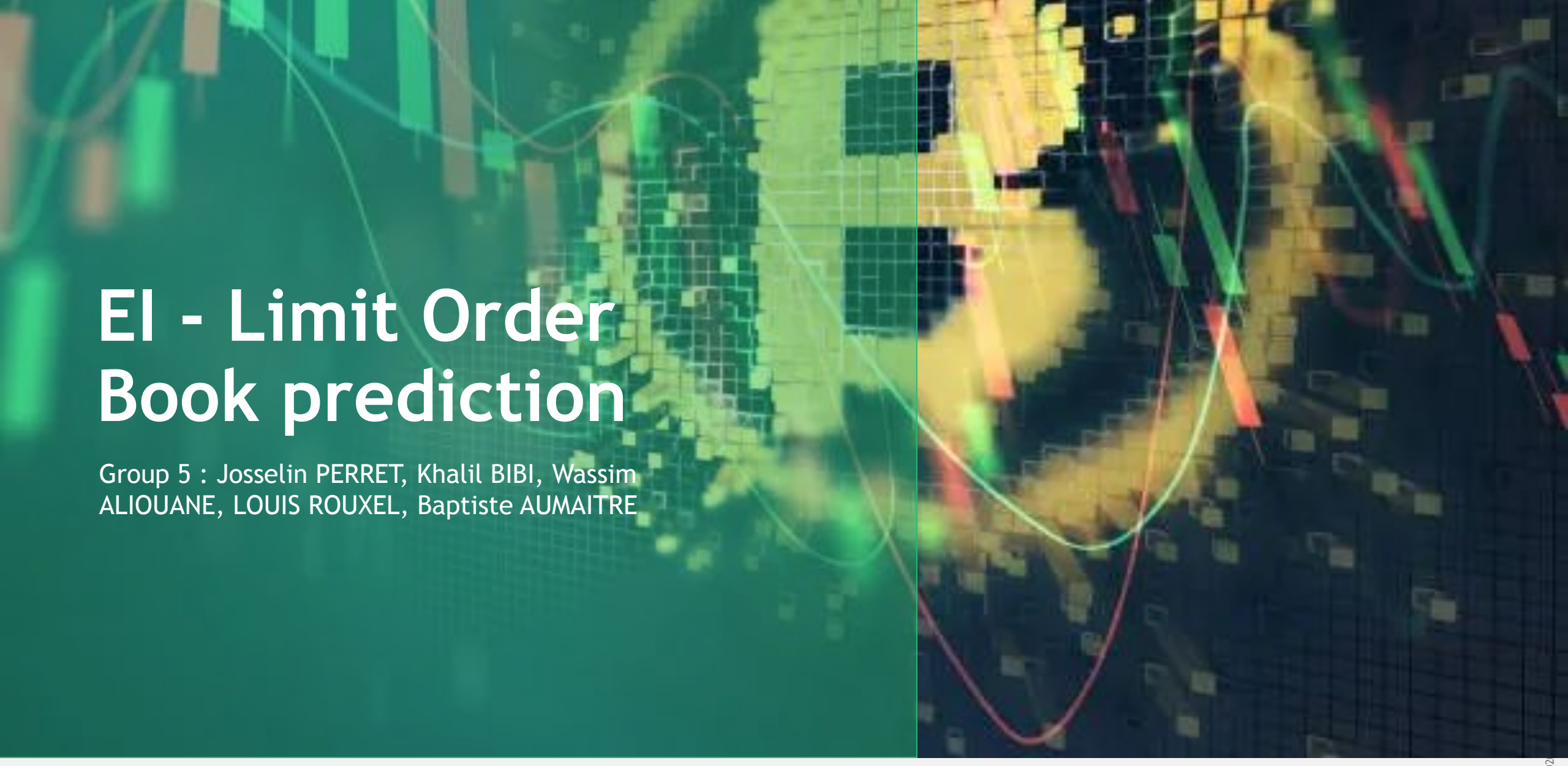


EI - Limit Order Book prediction

Group 5 : Josselin PERRET, Khalil BIBI, Wassim ALIOUANE, LOUIS ROUXEL, Baptiste AUMAITRE

CentraleSupélec

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Overview



Introduce the objectives

Analysis and processing the data

Features selection

Presentation of two strategies and how to assess the strategies performance

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Objectives of this study

- Study limit order books in the context of asynchronous order book data
- Design a cross-crypto strategy to predict the increase of ETH when influenced by other cryptocurrencies (Especially XBT)
- Backtest this strategy and assess the out-of-sample performances

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Processing of the data :

Large volume of data how to process ?



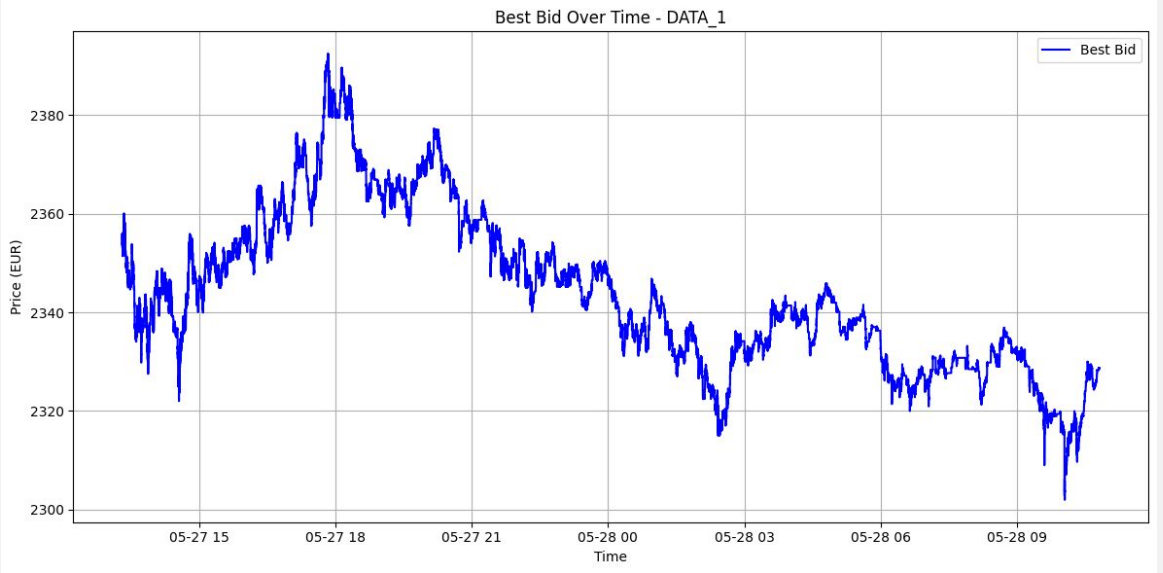
.csv → .parquet

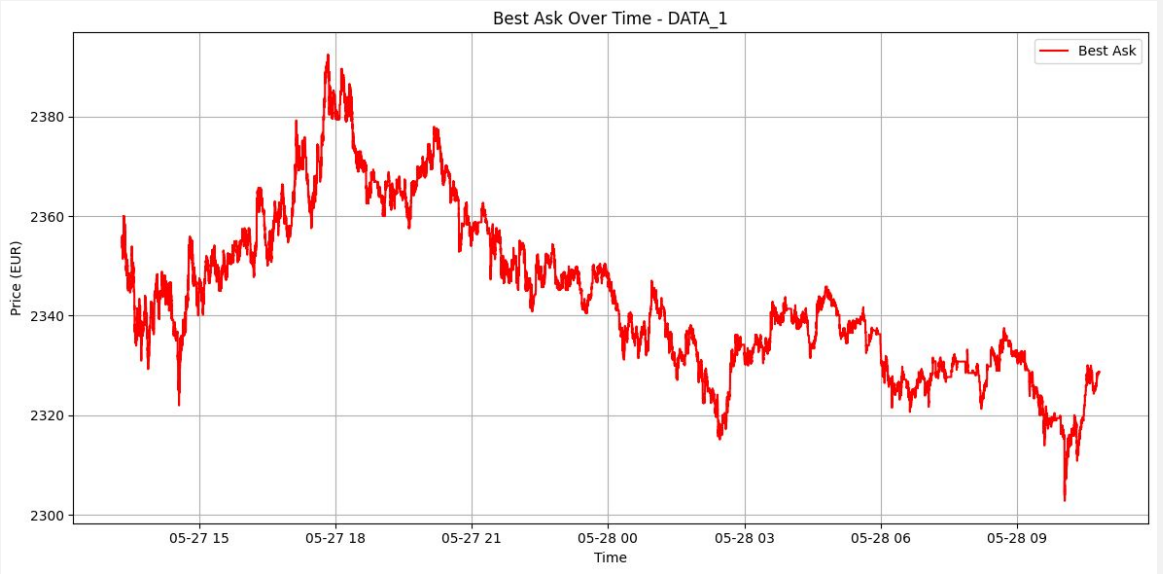
Calculate only once by saving snapshots in .parquet

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Preprocessing of the data

Extracting the best bid and best ask for each coin we are interested in

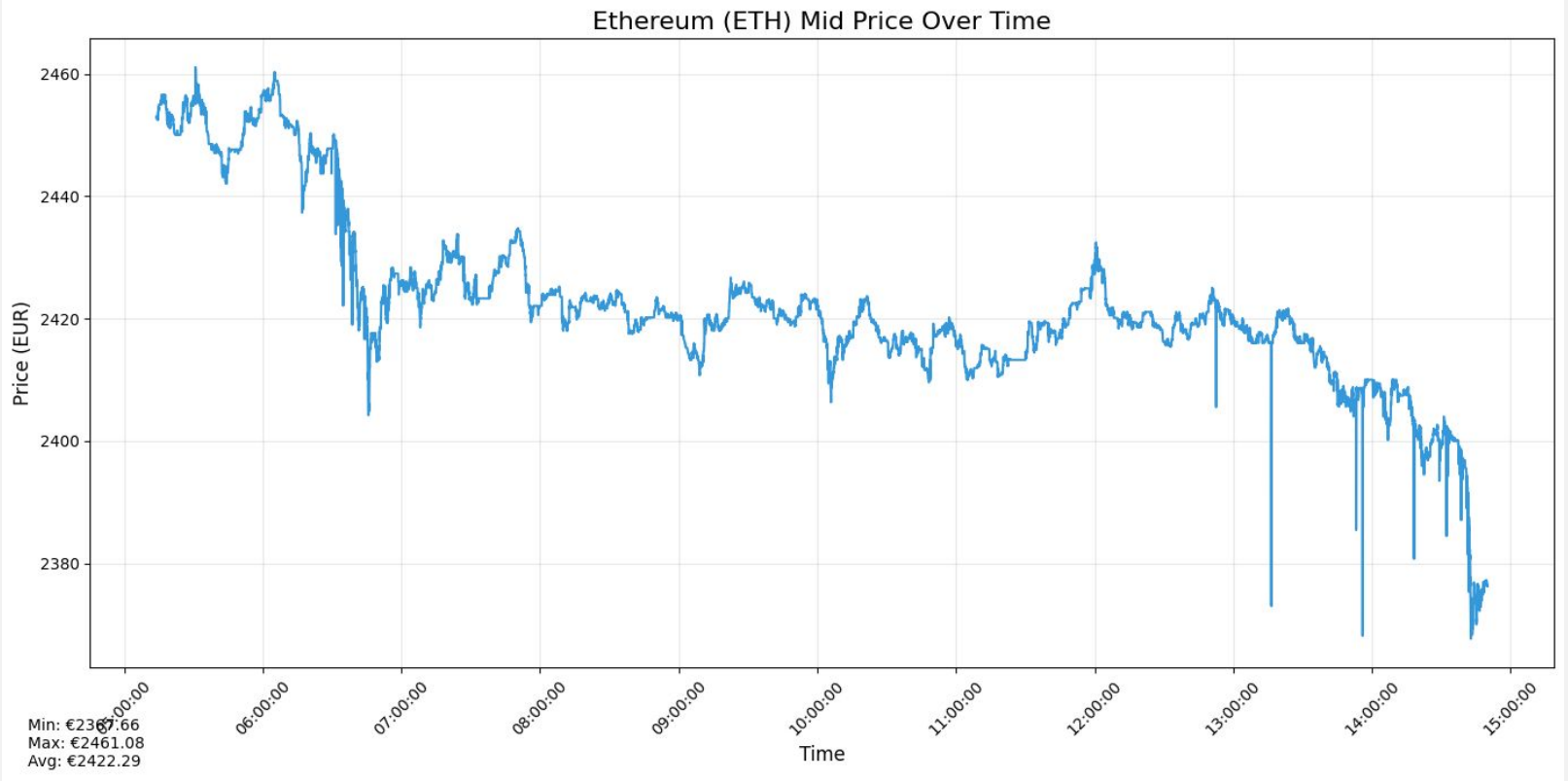




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Which signal we want to study :

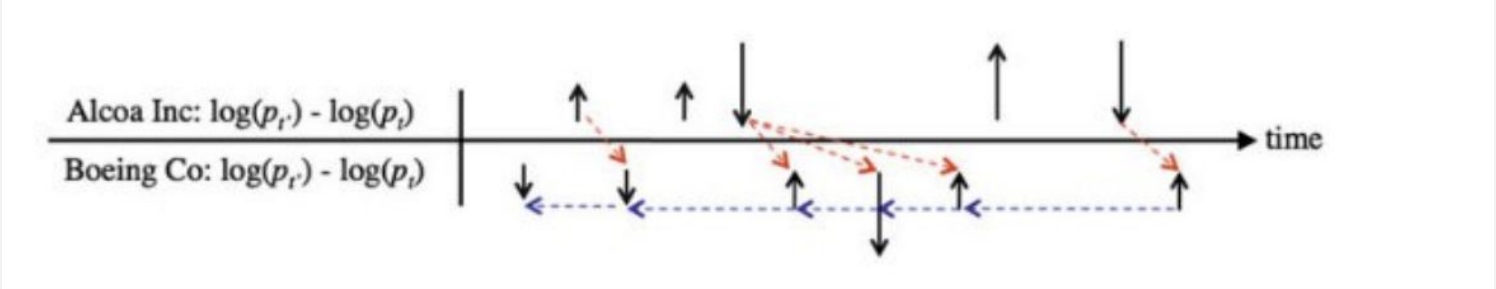
Mid price of ETH



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Alignment of XBT/ETH data

Difficulties in cross-crypto analysis :



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Features selection

What should we take into account in our model ?

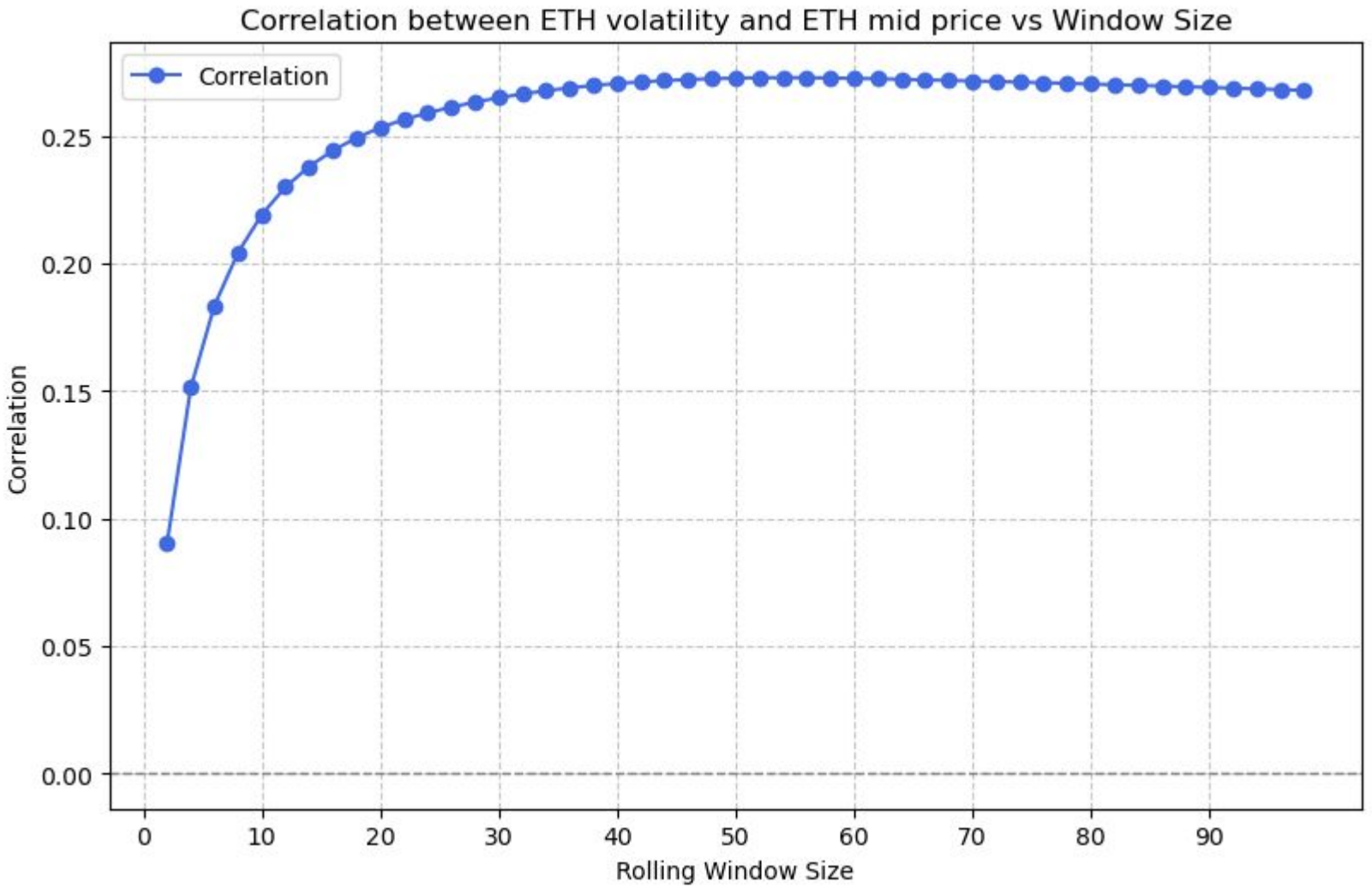
1. Lead lag optimization : Find the optimal time lag between variables to improve predictive power

2. First Statistical inference : Correlation matrix to quickly identify potential linear relationships between variables

3. Formal Causality Testing : Statistically test if one variable causes another in a predictive sense.

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Lead lag optimization



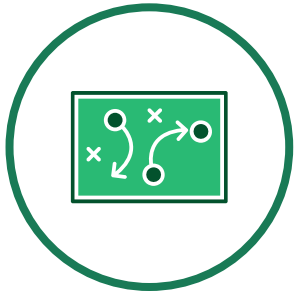
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List of features :



Price based features

- Mid-prices for ETH and BTC
- ETH/BTC price ratio
- Price spreads (absolute and percentage)



Volume-based features

- Total volumes for ETH and BTC
- Volume imbalances (bid vs ask)



Time-series features

- Price percentage changes
- Price volatility
- Moving averages for prices
- Volume moving averages
- Imbalance moving averages

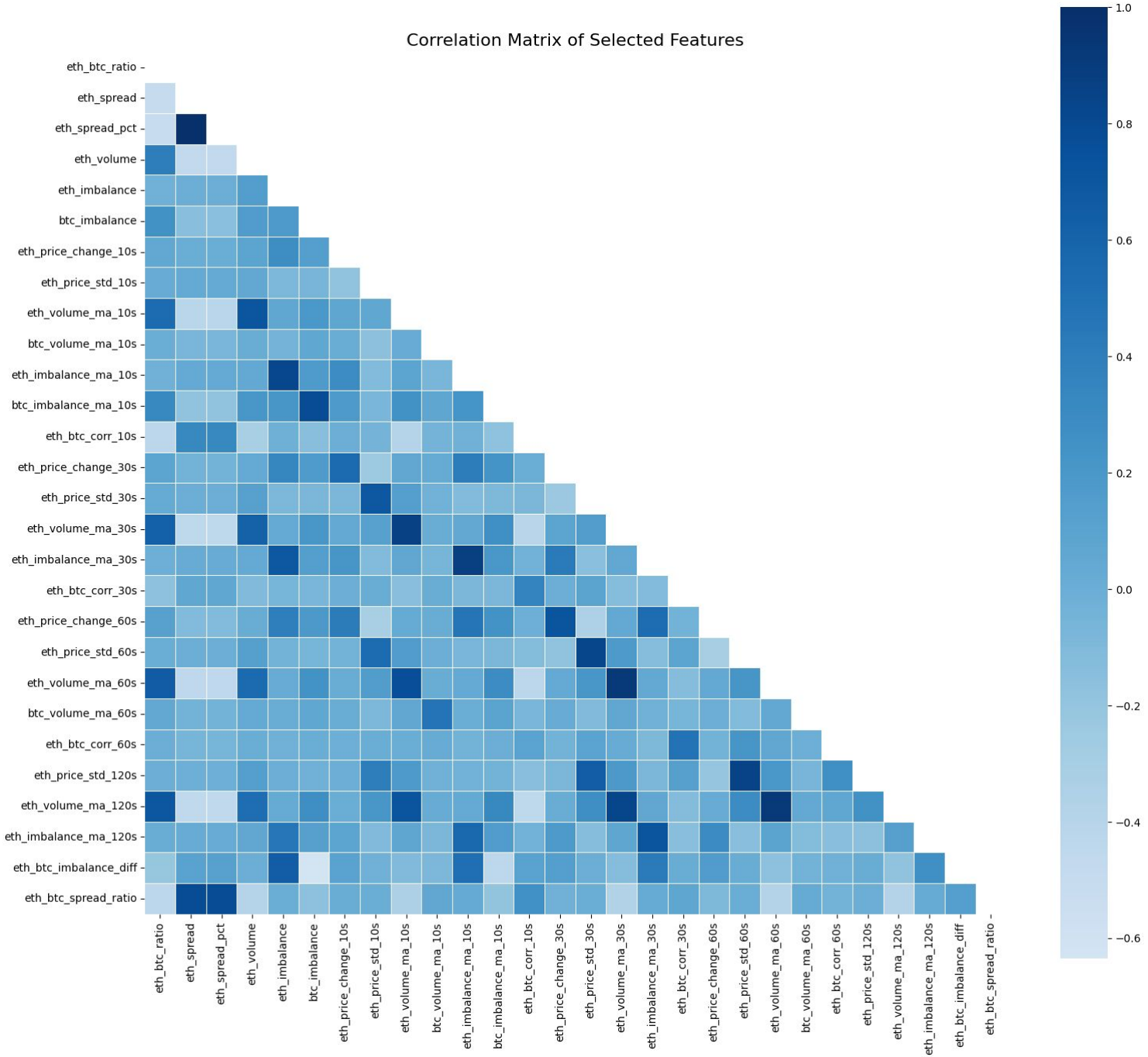


Cross-asset features

- ETH-BTC price correlations
- Imbalance differences
- Spread ratios

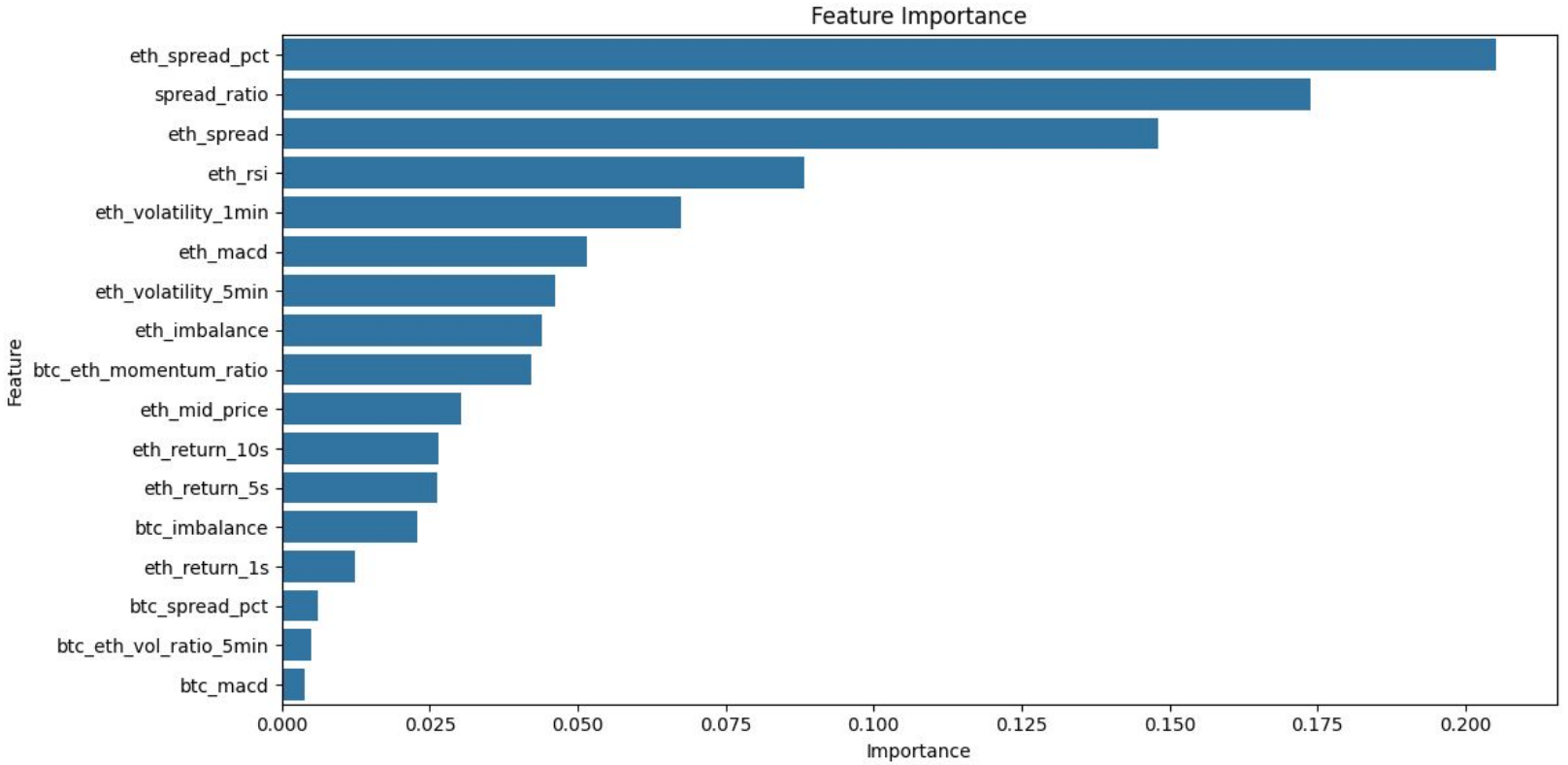
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Correlation Matrix



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Causality testing RandomForest :



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Key Findings (1)

Short-term Volatility Dominates

The top predictive features are short-term price volatility indicators:

- eth_price_std_10s (10-second price standard deviation)
- eth_price_change_10s (10-second price percentage change)
- eth_spread and eth_spread_pct (absolute and relative bid-ask spreads

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Key Findings (2)

Feature Selection :

Feature Selection: From the original 56 features, 28 were selected based on importance, reducing dimensionality while maintaining predictive power.

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To uncover real cross-crypto predictive signals :

For each candidate pair (BTC feature, ETH -feature), and each lag (delay) : measure of the TE from BTC to ETH

We looped over many feature combinations and lags on the entire in-sample DATASET0

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btc_feature	eth_feature	lag	te	p_value
return_10_symbol	imbalance_symbol	1	0.001482	0.0
return_10_symbol	imbalance_symbol	2	0.001480	0.0
return_20_symbol	return_20_symbol	2	0.001479	0.0
return_20_symbol	return_20_symbol	3	0.001467	0.0
return_20_symbol	return_20_symbol	1	0.001460	0.0
return_20_symbol	imbalance_symbol	2	0.001447	0.0
return_10_symbol	imbalance_symbol	3	0.001434	0.0
return_10_symbol	imbalance_symbol	4	0.001433	0.0
return_20_symbol	return_10_symbol	2	0.001424	0.0
return_20_symbol	return_20_symbol	4	0.001419	0.0
return_20_symbol	imbalance_symbol	1	0.001418	0.0
return_20_symbol	return_10_symbol	1	0.001418	0.0
return_10_symbol	return_20_symbol	2	0.001416	0.0
return_20_symbol	imbalance_symbol	3	0.001411	0.0
return_20_symbol	return_10_symbol	3	0.001409	0.0
return_10_symbol	return_20_symbol	1	0.001392	0.0

Causality Testing with transfer Entropy based analysis

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Strategy



Transfer Entropy based Strategy :

Return_10_symbol :
midprice(t)-midprice-(t-10)

Cross-crypto lead-lag signal : BTC Return_10_Symbol and ETH imbalance

Question : Is it a noise or a real signal ?

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Realistic backtest :
OOS DATASET 1

Based on the TE result :

Simple actionable signal

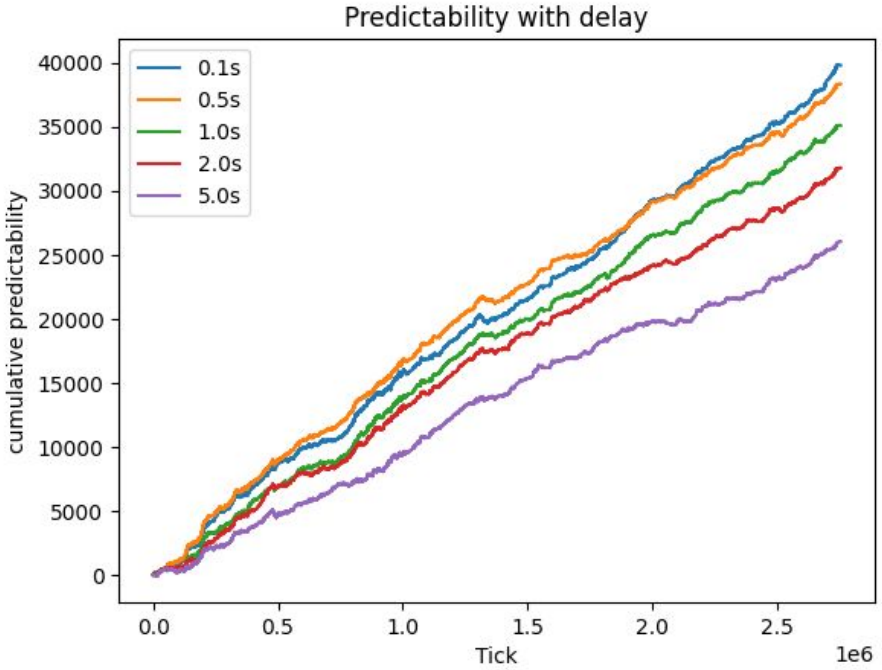
BTC UP -> Long ETH

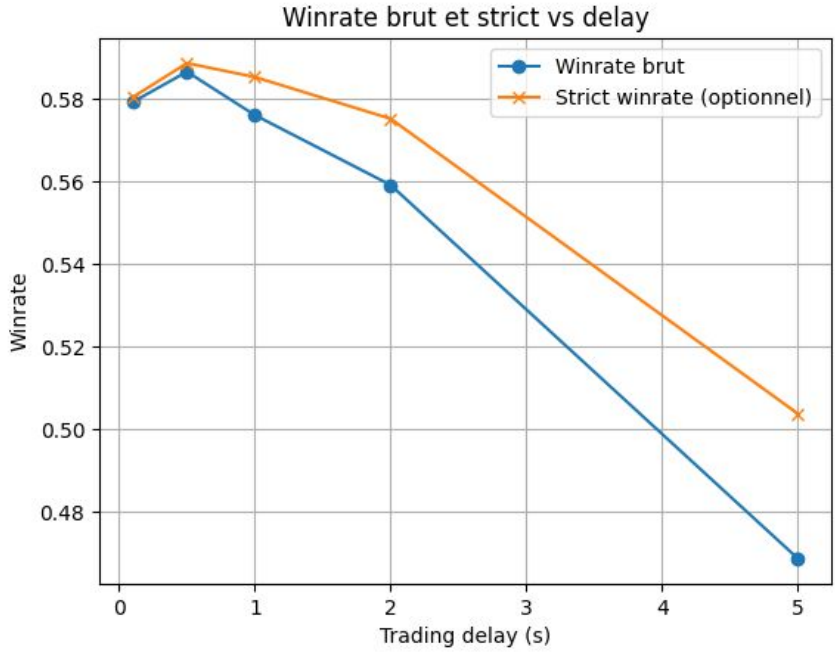
BTC DOWN -> Short ETH

BTC FLAT -> No position

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$$P = \sum_k a_k \quad \text{with} \quad a_k = \begin{cases} 1 & \text{if the prediction is correct} \\ -1 & \text{if the prediction is incorrect} \\ 0 & \text{if the prediction is indeterminate} \end{cases}$$





Impact of transaction-delay on performance :

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Strategy :

Trading Rule :

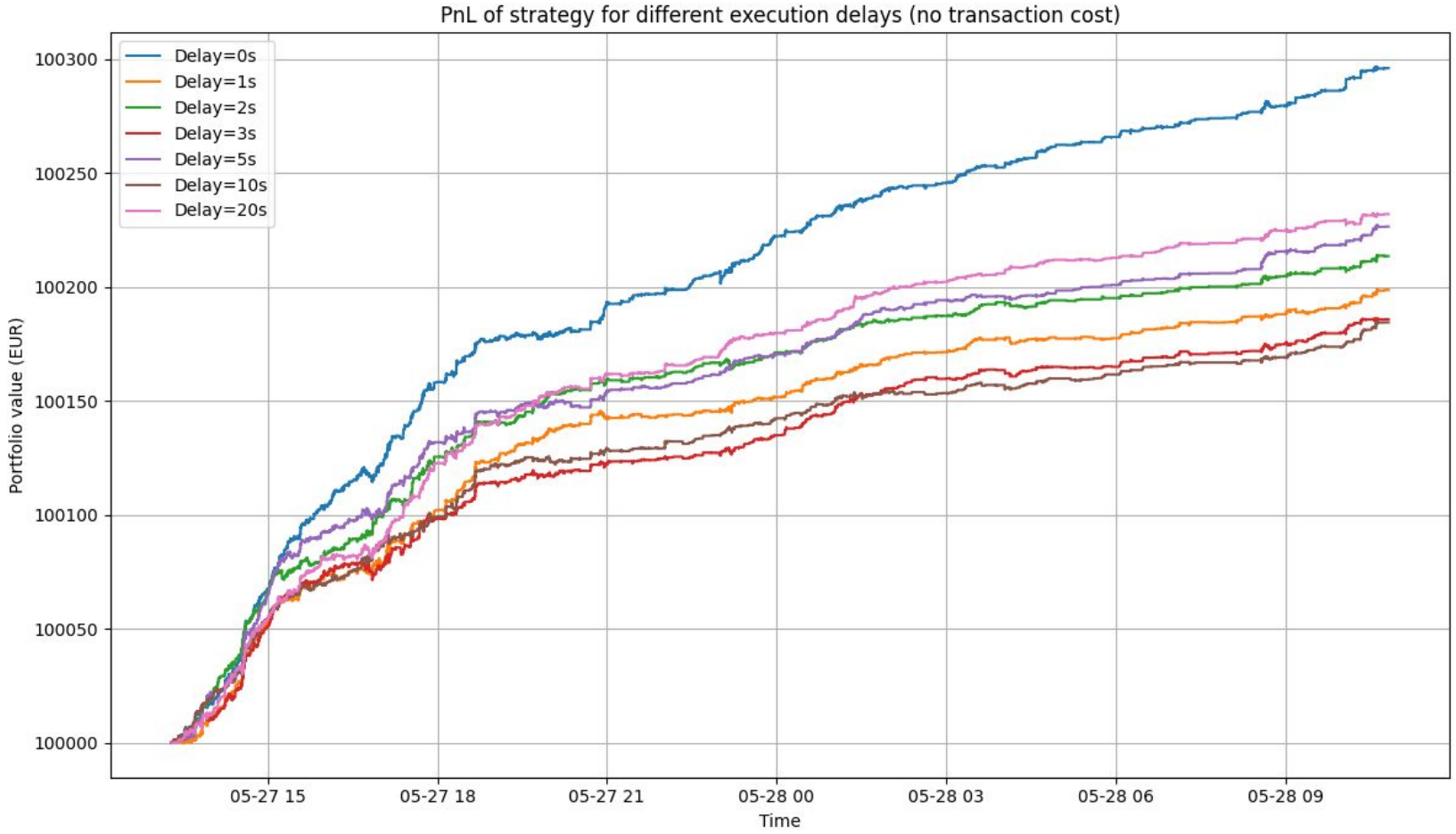
- Go LONG ETH when BTC UP
- Go SHORT ETH when BTC DOWN
- Stay FLAT when BTC FLAT
- EXIT/REVERSE the position when the signal changes

Portfolio Management :

- Fixed amount of Cash (100_000)
- Profits and losses : realized as the position is closed/reversed
- Portfolio value marked to market at every step
- The strategy does not compound or leverage positions each trade uses a fixed notional size

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Without exchange fees :



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Including realistic transaction costs

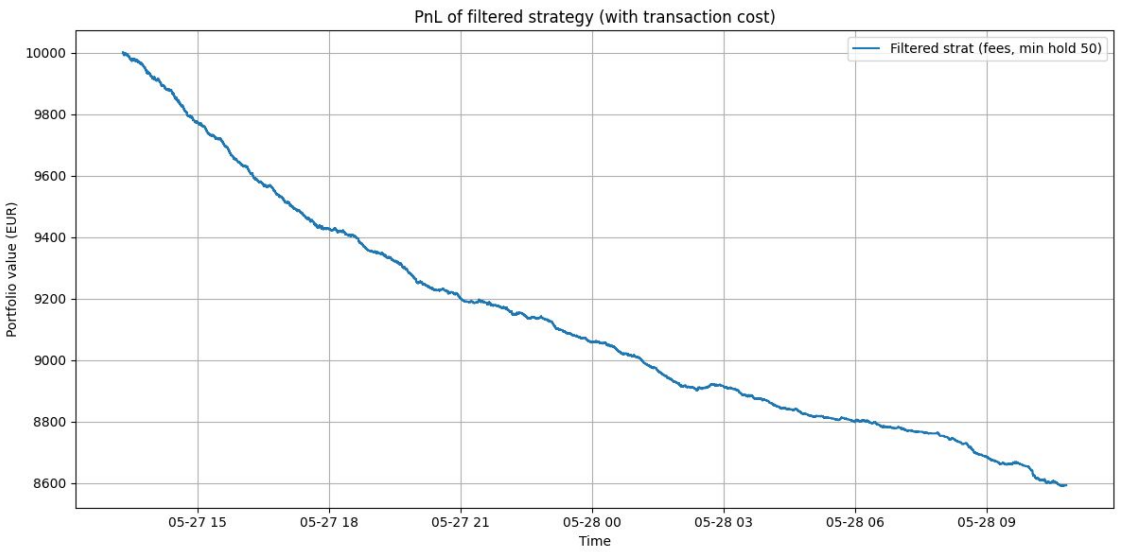
We evaluate the strategy using these assumptions:

- **Exchange fee**: 0.2% per trade
- **Slippage**: 0.1% per execution
- **Execution delay**: 1 second

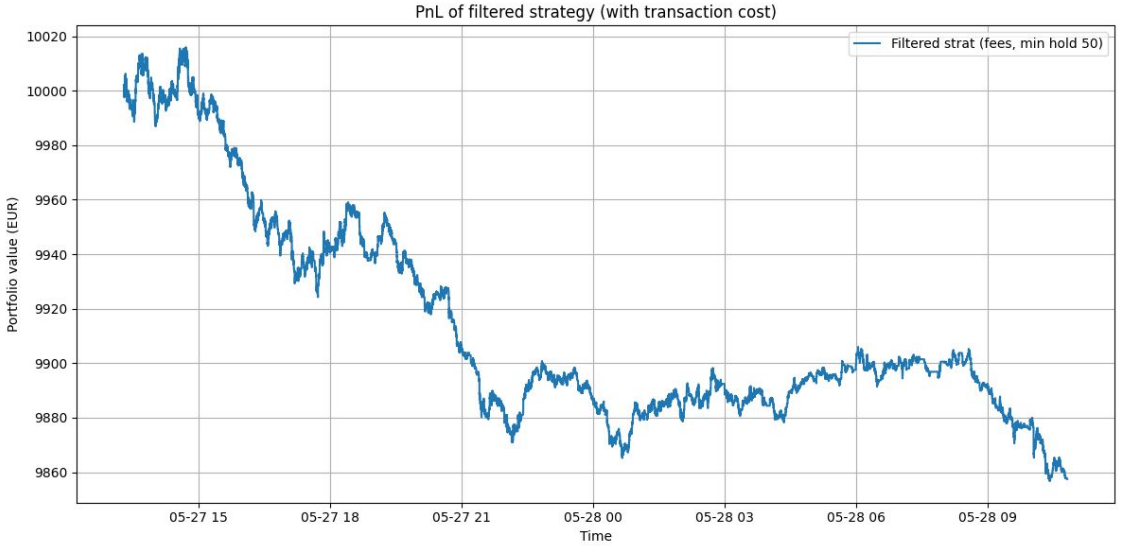
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Including realistic transaction costs



1 tick = 0,0005s



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Including realistic transaction costs



1 millions tick hold = 500s : not relevant for HFT

Total Brut perf: 0,5973%
Total Backtest duration : 21,52h
Alpha net/h : 2,78 bps/h

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Interpretation

The initial signal :

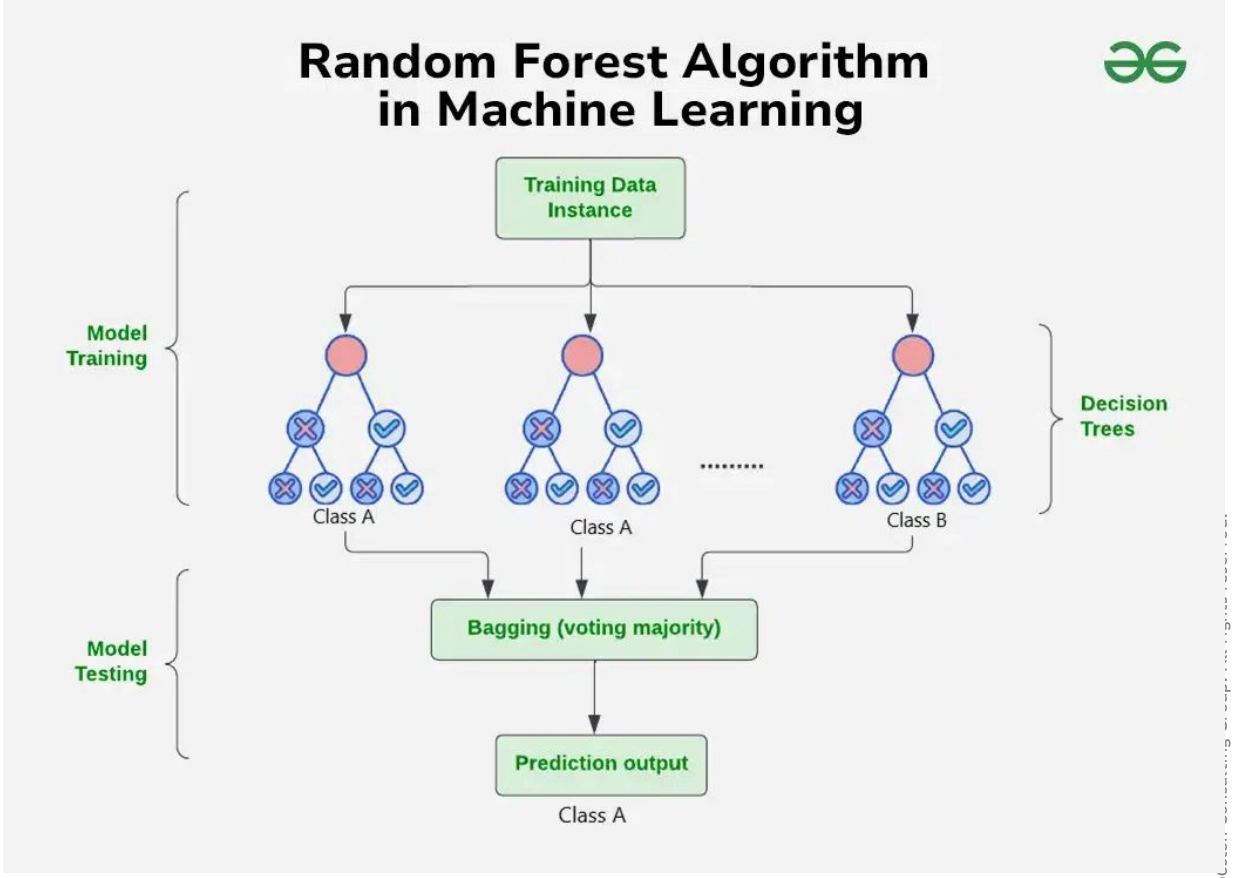
- Statistically significant TE
- A statistically significant signal is not necessarily exploitable after costs

Future work :

- Maybe the model is too simple (using only 2 features)
- Random forest based strategy

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Random Forest based Strategy



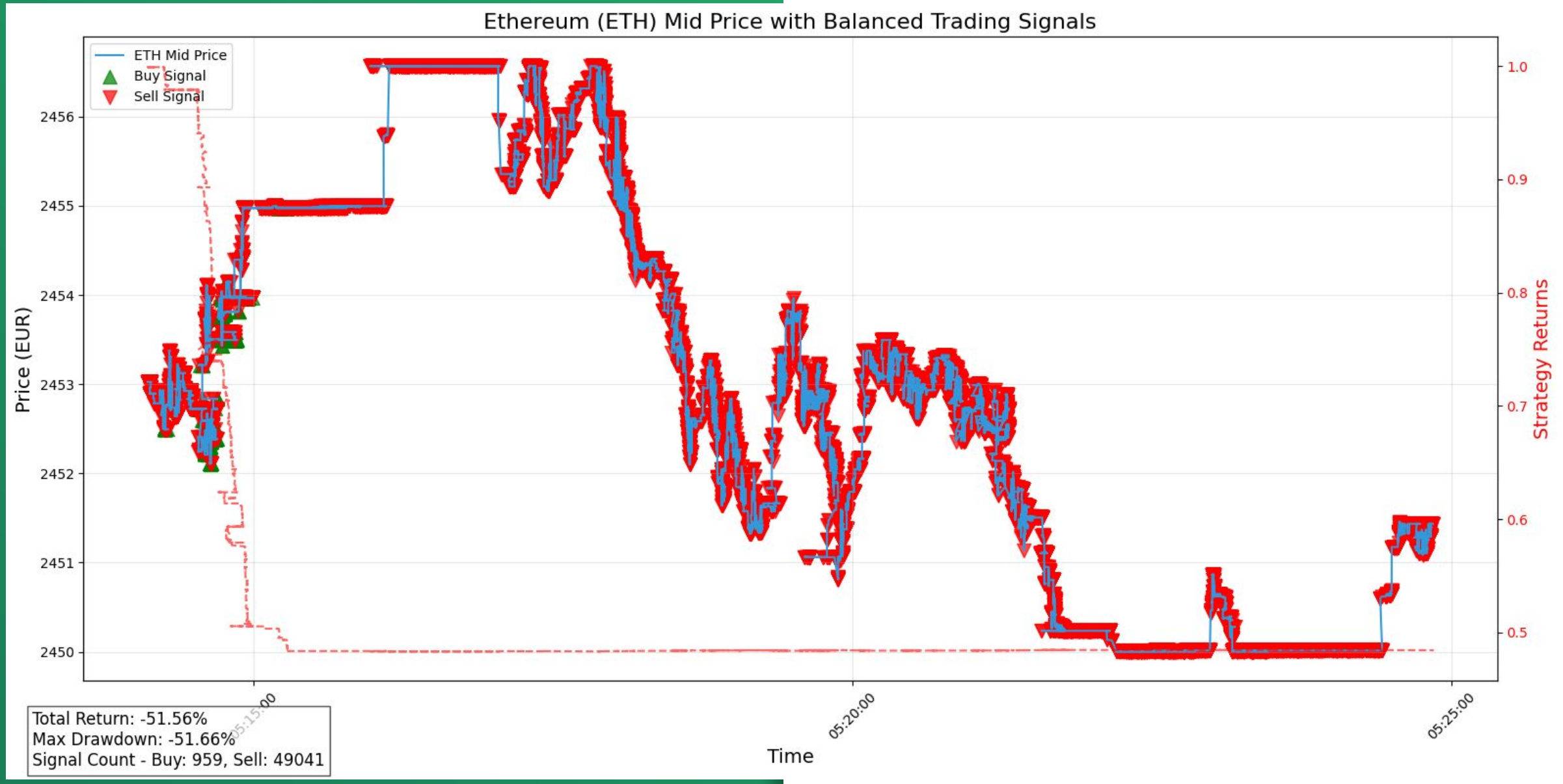
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Training of the model on 70% of DATASET_0 and testing on the last 30%.



Backtest on DATASET_1 and DATASET_2 to assess performances

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Biased data : The price of ETH is decreasing

```
Class Distribution in Training Set: [10544   546]
Majority Class Ratio: 0.9508
```



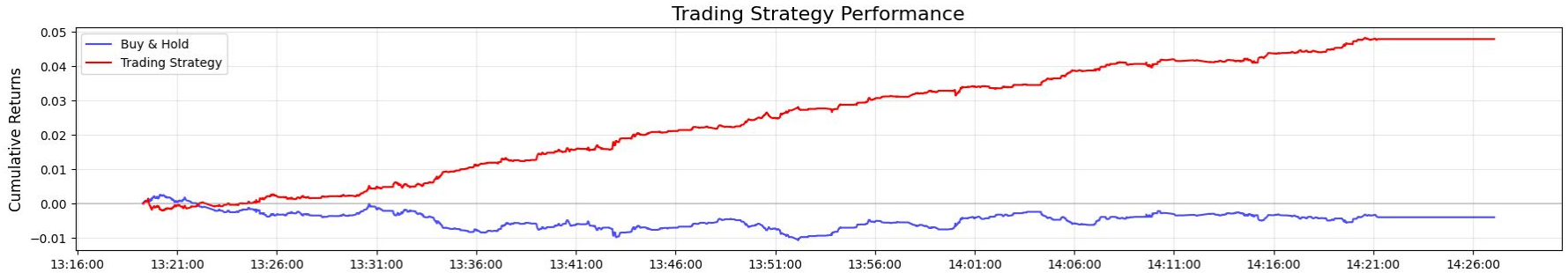
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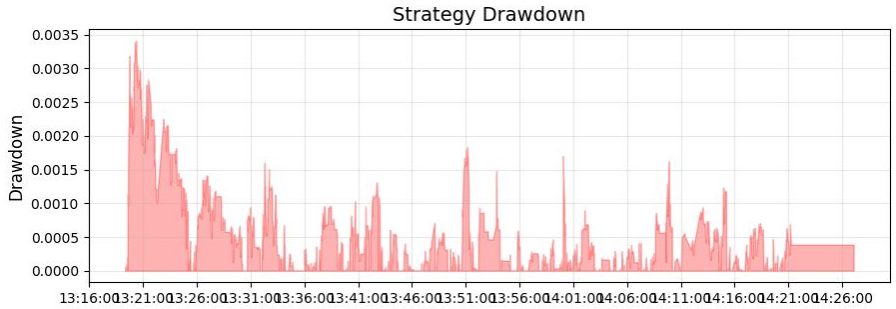
Weighted data included
Training window enlarged

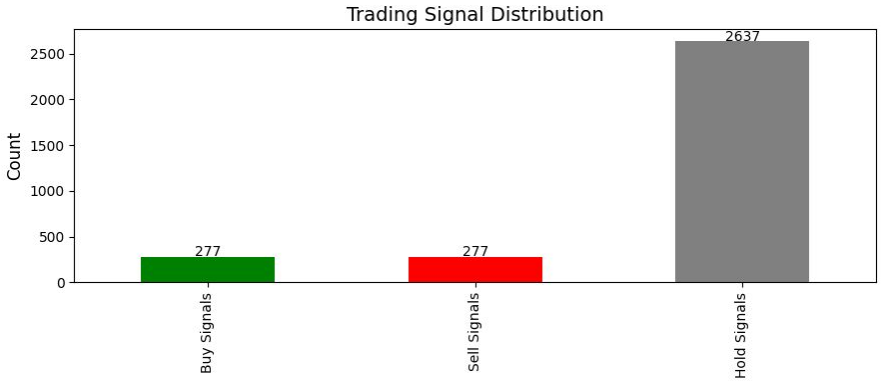


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Performance Evaluation on DATASET_1 (Without fees and without latency on transactions):



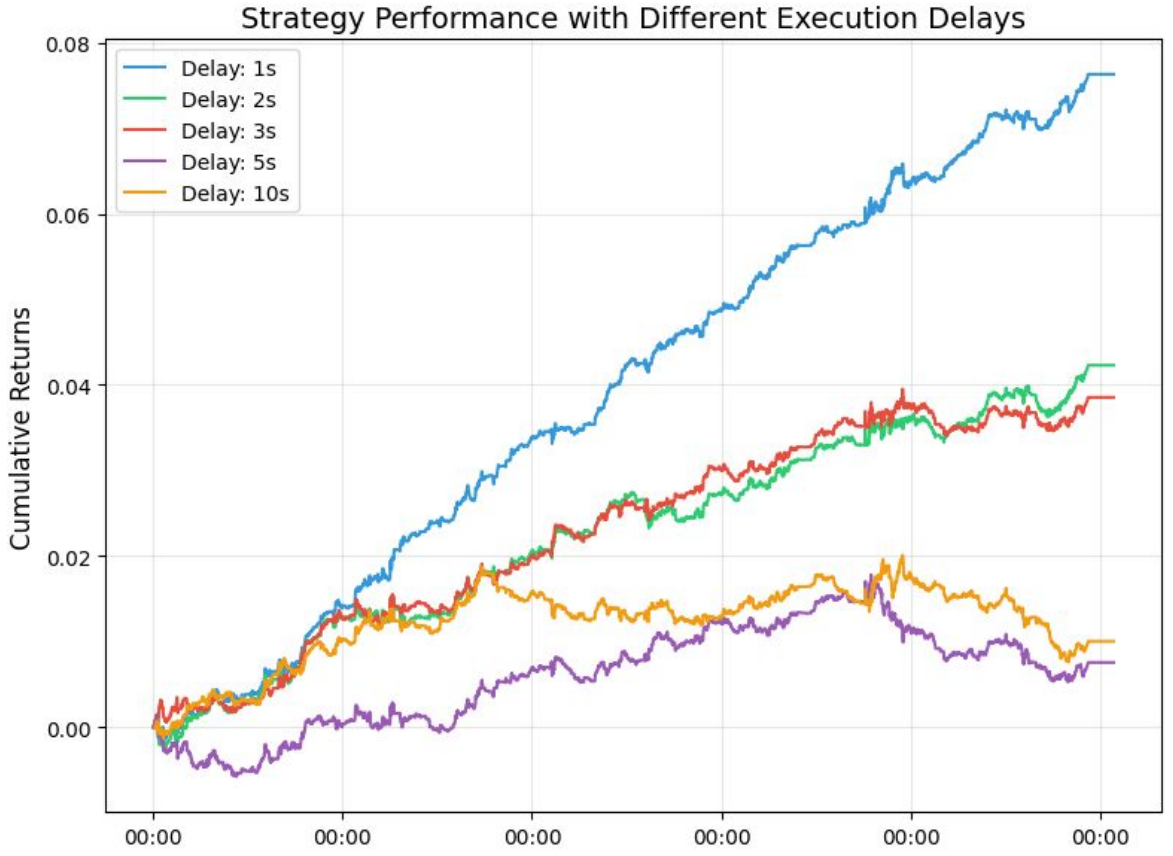






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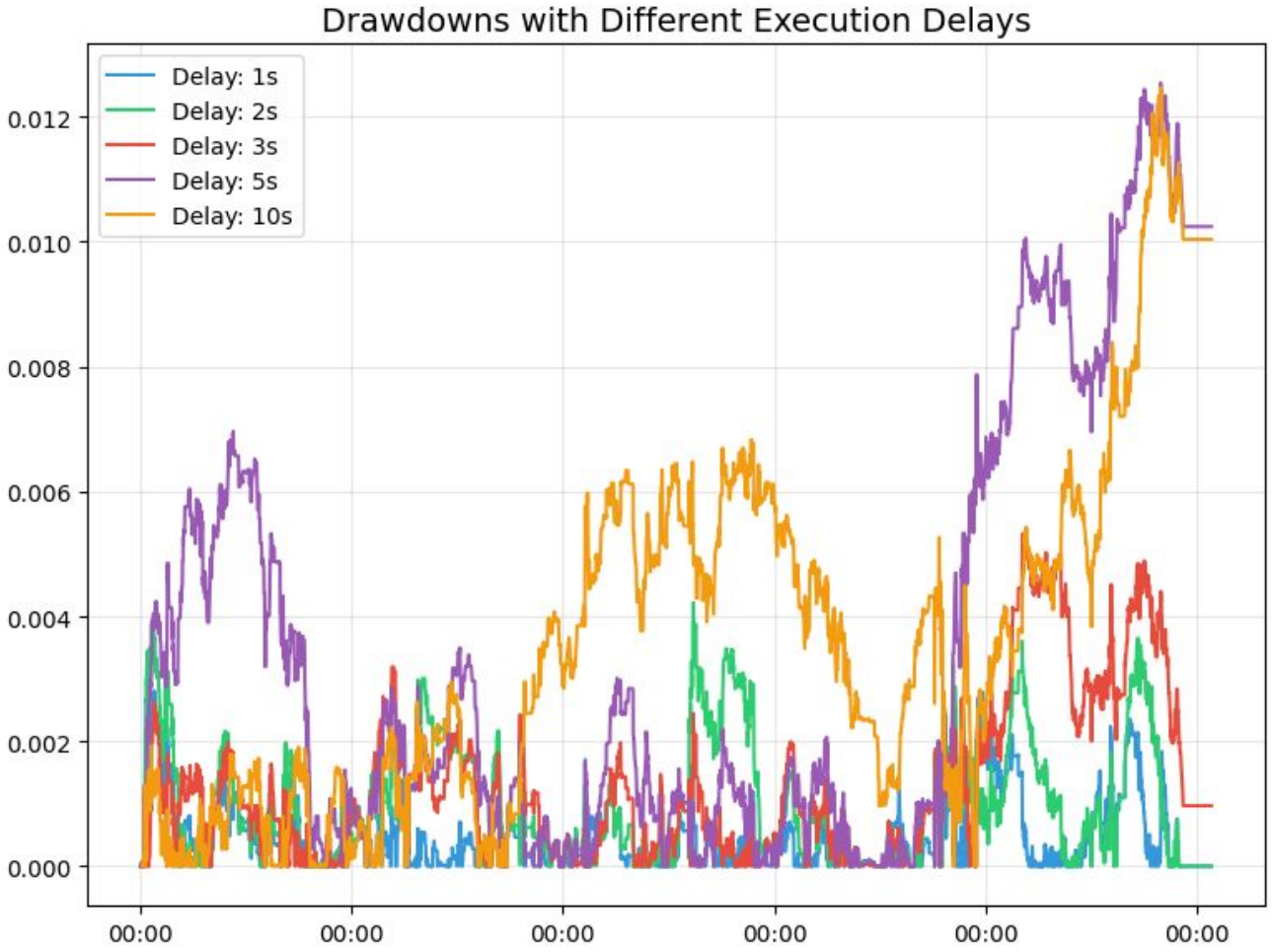
Impact of delay on model without fees (1)



Delay (seconds)	Total Return (%)
1	7.629871
2	4.229777
3	3.854760
5	0.757435
10	1.004375

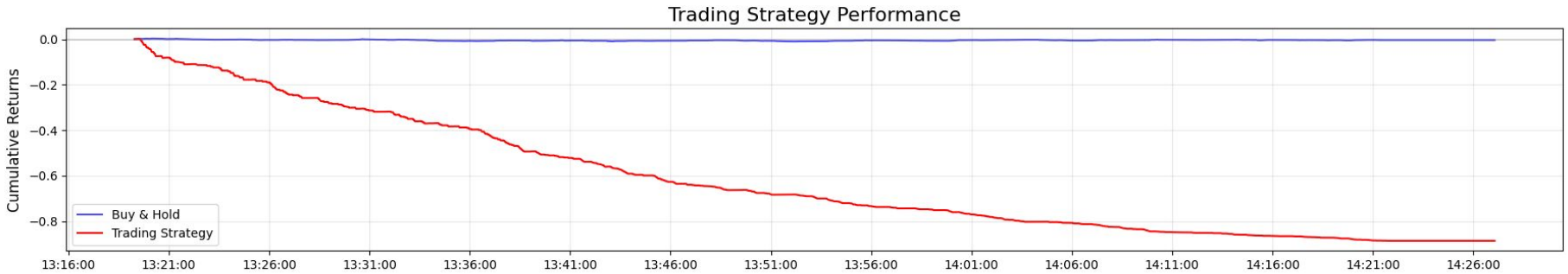
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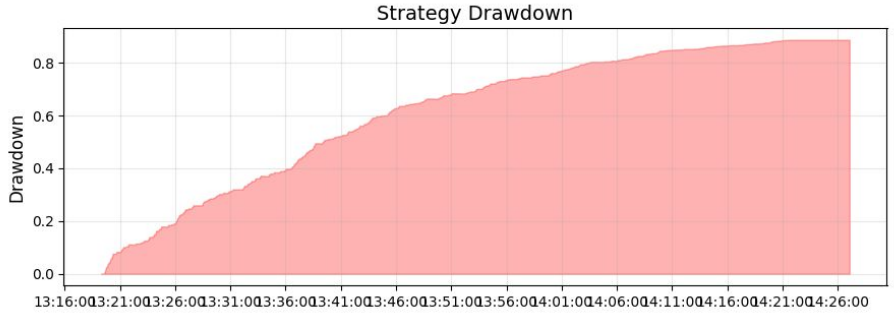
Impact of delay on model without fees (2)

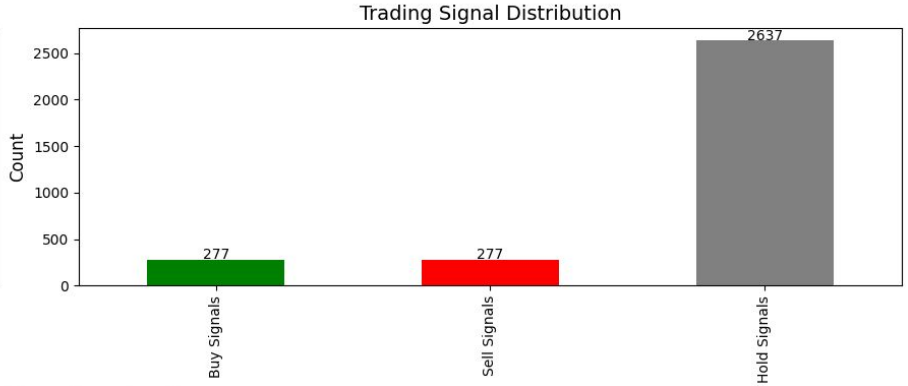


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Performance Evaluation on DATASET_1 (With transaction fees and 1 second latency):



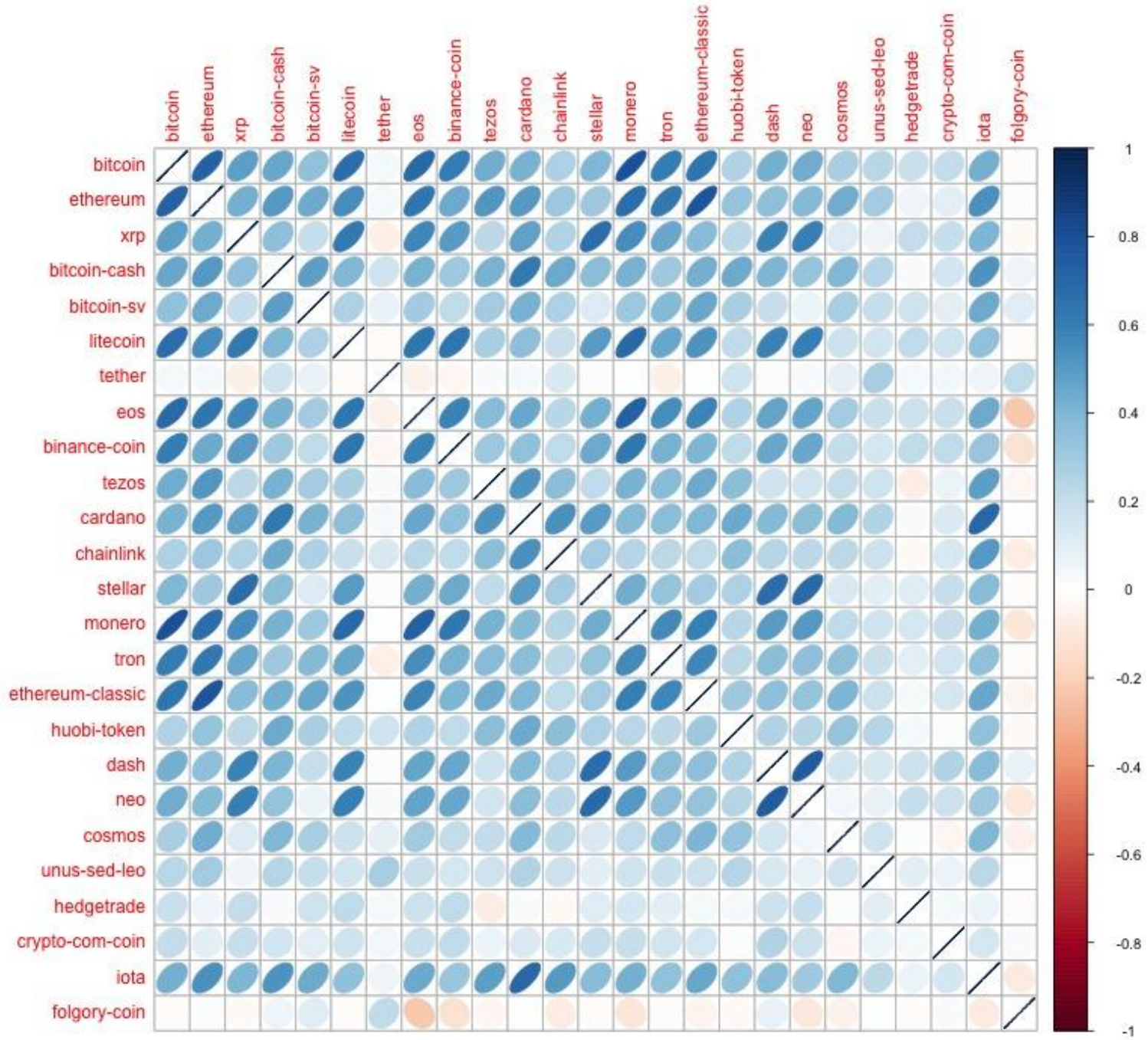






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Possible optimization



Considering others coins :

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Possible future optimizations :

- Take into consideration real-time delay on transactions (which depends on volumes on market orders)
- Train and test more models based on XGBOOST, LightGBM…



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Thanks for listening.

Why does both give different results ?

Transfer Entropy

- Is more concerned with causality than Random Forest (“"Does past X predict future Y better than past Y itself?")
- Gives the direction of the causality

Random Forest

- More about prediction than causality (“If we add X to our model, does it improve predictions of Y”)
- Helps determine how much each feature is contributing to the prediction of Y

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